Alexander Yao

(908) 381-9450 alex.r.yao@gmail.com Raritan, NJ
GitHub LinkedIn Website

EDUCATION

University of Maryland, College Park **College Park, MD**
B.S. in Computer Science Expected Dec. 2026
Minor in Applied Mathematics

- **Relevant Coursework**: Algorithms, Compilers, Advanced Data Structures, Computer Systems, Programming Languages, Web Development, Parallel Computing, Probability and Statistics, Computer and Network Security

SKILLS

Languages: Python, Java, Rust, C, OCaml, JavaScript, TypeScript, SQL, Racket, CSS, HTML
Systems & Backend: Linux, Networking, Async I/O, QUIC, Compilers, Data Pipelines, Android Studio
Tools: Git, Pandas, NumPy, Playwright, Node.js, Microsoft Azure, AWS, Vite

EXPERIENCE

George Mason University **Fairfax, VA**
Computer Vision & Security Research Assistant May 2025 – Present

- Designed and implemented a low-latency, cloud-native media delivery system over QUIC, achieving <150ms end-to-end latency.
- Integrated OpenCV-based media processing with QUIC transport to evaluate real-time video delivery under adversarial network conditions.
- Improved system reliability by 30% through packet scheduling, flow control tuning, and congestion simulation.
- Developed security testing modules to audit QUIC handshake and encryption behavior, strengthening transport-layer robustness.

Future Scientist Exchange Program **Hefei, China**
AI Security Research Intern June 2025 – August 2025

- Designed and executed controlled experiments simulating instruction-based and clean-label backdoor attacks against large language models (LLMs).
- Developed Python evaluation pipelines to quantify model evasion patterns and success rates using statistical robustness metrics.
- Presented technical findings to an international academic audience of 200+ attendees.

PROJECTS

Personal Portfolio & Media Gallery
React, TypeScript, Vite, CSS, Git Jan. 2026

- Engineered a high-performance portfolio using React Router v7 and TypeScript, implementing a centralized design system with native light/dark mode support via CSS variables.
- Designed a responsive media gallery component that optimizes performance by lazy-loading embeds and providing custom loading states for unlisted video content.
- Automated the deployment pipeline using GitHub Actions to build and host the production application on GitHub Pages upon every main-branch push.

Cross-Platform Video Performance Tracker
Python, SQL, Pandas Dec. 2025

- Designed a cross-platform data ingestion pipeline aggregating engagement metrics from YouTube and TikTok APIs.
- Implemented a headless browser scraping system using Playwright with session persistence and anti-bot mitigation to ensure reliable data collection.
- Developed platform-agnostic engagement normalization metrics (likes + comments per view) to enable fair, data-driven comparison across platforms.
- Transformed raw event data into clean, queryable analytics tables to support performance analysis.

Loot+ Systems Programming Project
Racket, x86-64 Assembly May 2025

- Implemented a Racket-to-x86-64 compiler supporting closures, variadic functions, and dynamic arity checks.
- Built a custom exception-handling mechanism via stack unwinding and non-local jumps using low-level register manipulation.
- Developed a full compiler pipeline (parsing, IR generation, codegen), achieving 100% behavioral parity with an interpreted reference model.